

A rigorously characterized I_{3322} quantum wall and finite-dimensional nonattainment

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Abstract

We determine the supremum of the I_{3322} Bell functional in both the tensor-product and commuting-operator models. The value is the rigorously characterized domain-wall constant

$$q_* = 0.250875384513976536\dots, \quad q_* \in [0.250875384513976535514, 0.250875384513976536486].$$

A computer-assisted shooting argument produces a positive Bellman fixed point and finite tensor-product strategies approaching q_* . A representation-free operator certificate, obtained by geometrically combining a Bellman inequality with its reflected counterpart, gives the matching commuting-operator upper bound. Equality on a finite joint spectrum forces two local decreasing transports to coincide. Their amplitude compatibility then bounds every putative finite attaining component by $1/4$. Since $q_* > 1/4$, no finite-dimensional tensor-product or commuting-operator strategy attains the supremum. The displayed decimal is a validated shooting zero, not an asserted elementary closed form. As a corollary, the finite-dimensional quantum correlation set in the three-input binary scenario is not closed; this is the minimal bipartite binary-output scenario by input counts where nonclosure can occur.

1 Statement of the result

Let A_1, A_2, A_3 and B_1, B_2, B_3 be projections. In the commuting model they act on one Hilbert space and $[A_i, B_j] = 0$. We use the normalization

$$\begin{aligned} \mathcal{B}_{3322} = & -A_2 - B_1 - 2B_2 + A_1B_1 + A_1B_2 + A_2B_1 + A_2B_2 \\ & - A_1B_3 + A_2B_3 - A_3B_1 + A_3B_2. \end{aligned} \tag{1}$$

This is the inequality introduced by Collins and Gisin [3]. Its classical bound is zero. Qubit strategies attain $1/4$. Pál and Vértesi constructed increasing-dimensional strategies approaching approximately 0.250875384514 and conjectured finite-dimensional nonattainment [9]. Our projector convention agrees term-for-term with their best-response recursion. In the standard Collins–Gisin table it is obtained by swapping the first two settings on both sides; the exact affine conversion to the dichotomic convention is recorded in the accompanying normalization concordance.

Theorem 1. *There is a rigorously characterized constant q_* satisfying*

$$q_* \in [0.250875384513976535514, 0.250875384513976536486] \tag{2}$$

for which

$$\omega_{\otimes}(\mathcal{B}_{3322}) = \omega_c(\mathcal{B}_{3322}) = q_*.$$

No finite-dimensional tensor-product strategy and no finite-dimensional commuting-operator strategy attains q_ .*

The NPA hierarchy [8] and its recent ncKKT refinements [1] provide increasingly sharp numerical bounds for this problem. Our proof instead extracts an exact operator certificate from the limiting domain-wall recursion. Infinite-versus-finite dimensional separations and non-closure are known in larger Bell scenarios [11, 2]; the point here is the exact resolution of the canonical three-setting binary functional. Mghirbi previously gave proof-carrying exact rational upper and lower certificates enclosing the value to width below 10^{-9} [7] (doi:10.5281/zenodo.21477901). That important antecedent explicitly left the exact supremum, finite-dimensional attainment, and tensor-commuting equality open. Very recent budgeted moment-selection work continues to use I_{3322} as a hard NPA benchmark rather than supplying an exact theorem [5].

1.1 Minimal binary nonclosure

Let $C_q(m_A, m_B; 2, 2)$ denote the set of finite-dimensional bipartite quantum correlations with m_A and m_B binary measurements. We allow POVMs; this is equivalent to the finite-dimensional PVM convention.

Theorem 2 (Minimal binary nonclosure). *The set $C_q(3, 3; 2, 2)$ is not closed. If either $m_A \leq 2$ or $m_B \leq 2$, then $C_q(m_A, m_B; 2, 2)$ is compact. Hence $(3, 3; 2, 2)$ is minimal, under the coordinatewise ordering of input counts, among bipartite binary-output scenarios with nonclosed finite-dimensional quantum correlation sets.*

Proof. The certified Bellman datum supplies finite-dimensional correlations p_n whose I_{3322} values converge to q_* . The probability vectors lie in a fixed compact cube, so a subsequence converges to some p_∞ in $\overline{C_q(3, 3; 2, 2)}$. Continuity gives $I_{3322}(p_\infty) = q_*$. Theorem 1 excludes every finite-dimensional attainer, so $p_\infty \notin C_q(3, 3; 2, 2)$. This proves nonclosure. The implication from I_{3322} nonattainment to this statement was explicitly anticipated by Dykema, Paulsen, and Prakash [4].

For minimality, consider $p \in C_q(2, m; 2, 2)$. Simultaneously dilate Alice’s two binary POVMs to two projections. Jordan’s lemma decomposes their common finite-dimensional space into invariant blocks of dimension at most two. Alice’s block-diagonal measurements make p a convex mixture of the corresponding block correlations. After purifying each block state into Bob’s system, Schmidt compression reduces Bob’s support to dimension at most two without changing any probability. Thus

$$C_q(2, m; 2, 2) = \text{conv}(K_m),$$

where K_m is the compact set of qubit–qubit behaviors with the prescribed measurements. The convex hull of a compact subset of a finite-dimensional space is compact. More explicitly, the no-signaling affine dimension is $3m + 2$, so Carathéodory reduces every behavior to at most $3m + 3$ qubit blocks, giving local dimensions at most $6m + 6$. The argument with the parties interchanged proves the other case. Related recent dimension bounds appear in [6, 10]. $\square$

The theorem is only a binary-output, bipartite setting-count statement. It does not compare scenarios with more outputs, more parties, communication, or network causal structure. It also does not decide whether an infinite-dimensional spatial strategy attains q_* .

2 Certified Bellman datum

Put

$$b(x) = \frac{\sqrt{1-x^2}}{2}, \quad d(x, u) = xu + \frac{x-u}{2} - 1. \quad (3)$$

The only computer-assisted input to the analytic proof is the following.

Proposition 3 (Certified Bellman datum). *There exist q_* satisfying (2), a positive continuous concave function $F : [-1, 1] \rightarrow (0, \infty)$, and a strictly increasing map $P : [-1, 1] \rightarrow [-1, 1]$ such that, with*

$$p(x) = \frac{b(x)^2}{F(x)},$$

one has

$$p(x) + F(u) \leq q_* - d(x, u) \quad (4)$$

for every $x, u \in [-1, 1]$. For each u , equality holds at the unique point $x = P(u)$. Moreover, a sequence of finite-dimensional tensor-product strategies has Bell values converging to q_* .

Equivalently, F is a fixed point of

$$(T_q F)(u) = \min_{x \in [-1, 1]} \left(q + 1 - \frac{x}{2} - (x - \frac{1}{2})u - \frac{1 - x^2}{4F(x)} \right). \quad (5)$$

The validated construction of Proposition 3 is summarized in Section 6.

3 Geometric symmetrization

Apply (4) at (x, u) and $(-u, -x)$. Since $d(-u, -x) = d(x, u)$,

$$\begin{aligned} p(x) + F(u) &\leq q_* - d(x, u), \\ F(-x) + p(-u) &\leq q_* - d(x, u). \end{aligned} \quad (6)$$

Define

$$a(x) = \sqrt{p(x)F(-x)}, \quad c(u) = \sqrt{F(u)p(-u)}. \quad (7)$$

Cauchy–Schwarz applied to $(\sqrt{p(x)}, \sqrt{F(u)})$ and $(\sqrt{F(-x)}, \sqrt{p(-u)})$ gives

$$a(x) + c(u) \leq q_* - d(x, u). \quad (8)$$

Direct multiplication yields

$$a(x)a(-x) = b(x)^2, \quad c(u)c(-u) = b(u)^2. \quad (9)$$

4 Commuting-operator certificate

Set

$$X = A_1 + A_2 - I, \quad Y = A_2 - A_1, \quad U = B_1 + B_2 - I, \quad V = B_2 - B_1. \quad (10)$$

The projection relations imply

$$X^2 + Y^2 = I, \quad XY + YX = 0, \quad U^2 + V^2 = I, \quad UV + VU = 0. \quad (11)$$

A direct expansion gives

$$\mathcal{B}_{3322} = G(X, U) + Y(B_3 - I/2) + (A_3 - I/2)V, \quad (12)$$

where

$$G(X, U) = XU + X/2 - U/2 - I.$$

Let

$$\alpha(x) = \frac{1}{2} - a(x), \quad \beta(u) = q_* - \frac{1}{2} - c(u).$$

Because X and U commute, joint functional calculus applied to (8) gives

$$R_0 := \alpha(X) + \beta(U) - G(X, U) \geq 0. \quad (13)$$

Let J_A be the polar sign of Y on the support of $I - X^2$. Then

$$Y = 2b(X)J_A, \quad J_A f(X) = f(-X)J_A. \quad (14)$$

Put $S_B = 2B_3 - I$ and $L_A = \sqrt{a(X)}$. Cross-party commutation gives $[J_A, S_B] = 0$. Using (9),

$$L_A J_A S_B L_A = b(X)J_A S_B = Y(B_3 - I/2). \quad (15)$$

Consequently

$$\begin{aligned} R_A &:= \frac{1}{2}I - \alpha(X) - Y(B_3 - I/2) \\ &= L_A(I - J_A S_B)L_A \geq 0. \end{aligned} \quad (16)$$

The product $J_A S_B$ is a self-adjoint contraction. Endpoint summands, where $b(X) = Y = 0$, follow by continuity. The identical Bob construction gives

$$R_B := (q_* - \frac{1}{2})I - \beta(U) - (A_3 - I/2)V = \sqrt{c(U)}(I - S_A J_B)\sqrt{c(U)} \geq 0. \quad (17)$$

Adding (13), (16), and (17) yields

$$\boxed{q_* I - \mathcal{B}_{3322} = R_0 + R_A + R_B \geq 0.} \quad (18)$$

Thus $\omega_c \leq q_*$. Proposition 3 supplies a tensor-product sequence approaching q_* , proving equality of both suprema.

5 Finite-dimensional nonattainment

Suppose a finite-dimensional strategy attains q_* , and let ψ be an attaining unit vector. Positivity in (18) implies

$$R_0 \psi = R_A \psi = R_B \psi = 0. \quad (19)$$

The scalar value of R_0 at a joint spectral pair (x, u) is $q_* - d(x, u) - a(x) - c(u)$. Equality forces equality in both inequalities of (6) and in Cauchy–Schwarz. Hence every occupied pair obeys

$$x = P(u), \quad -u = P(-x). \quad (20)$$

Let Σ be the finite set of occupied u -coordinates. Strict monotonicity makes P injective.

On the occupied interior support define

$$K_A = J_A S_B, \quad K_B = S_A J_B, \quad r_A(x) = \frac{a(x)}{b(x)}, \quad r_B(u) = \frac{c(u)}{b(u)}.$$

The local equations in (19) become

$$K_A \psi = r_A(X) \psi, \quad K_B \psi = r_B(U) \psi. \quad (21)$$

The first reverses x and therefore acts on Σ by

$$\mathfrak{a}(u) = P^{-1}(-P(u)), \quad (22)$$

while the second acts by

$$\mathfrak{b}(u) = -u. \quad (23)$$

Both are decreasing bijections of the same finite ordered set. Such a set has only one decreasing bijection, so $\mathfrak{a} = \mathfrak{b}$. Every occupied (x, u) is paired with $(-x, -u)$.

Let ψ_+ and ψ_- be the corresponding nonzero spectral components. Taking norms in (21) and using (9) gives

$$\rho := \frac{\|\psi_-\|}{\|\psi_+\|} = \sqrt{\frac{F(-x)}{F(x)}} = \sqrt{\frac{F(u)}{F(-u)}}. \quad (24)$$

Bellman and Cauchy equality then force

$$F(x) = \frac{b_x}{\rho}, \quad F(-x) = \rho b_x, \quad F(u) = \rho b_u, \quad F(-u) = \frac{b_u}{\rho}. \quad (25)$$

At the two reflected components,

$$q_* - d(x, u) = \rho(b_x + b_u), \quad q_* - d(-x, -u) = \frac{b_x + b_u}{\rho}. \quad (26)$$

Subtracting and eliminating ρ gives

$$q_* = xu - 1 + \sqrt{(b_x + b_u)^2 + \frac{(x - u)^2}{4}}. \quad (27)$$

Put $s_x = \sqrt{1 - x^2}$, $s_u = \sqrt{1 - u^2}$, and $t = 1 - xu \geq 0$. Then

$$(b_x + b_u)^2 + \frac{(x - u)^2}{4} = \frac{1 - xu + s_x s_u}{2} \leq t, \quad (28)$$

since

$$(1 - xu)^2 - s_x^2 s_u^2 = (x - u)^2 \geq 0.$$

Therefore

$$q_* \leq -t + \sqrt{t} \leq \frac{1}{4}, \quad (29)$$

contradicting (2). This proves finite-dimensional nonattainment. Independently, a finite-dimensional commuting representation decomposes into finitely many spatial matrix blocks; equality would force one finite tensor block to attain q_* .

6 Validated construction of the Bellman datum

The Bellman equality and stationarity equations close as a three-dimensional shooting map. In coordinates (z, x, r) , put

$$\Delta = zx + \frac{z - x}{2} - 1, \quad v = \frac{2(q - \Delta - \sqrt{1 - z^2}/(2r))}{\sqrt{1 - x^2}}.$$

The successor coordinate is the explicit rational-radical expression obtained from stationarity. The selected orbit leaves a hyperbolic positive plateau, meets a two-condition reflection section, and returns by reversing symmetry.

The validation stack comprises: a 400-bit Miranda enclosure of the shooting zero; a symbolic envelope-covariance identity; 300-bit interval covers of the central graph and endpoint wings; strict derivative signs on every tile; exclusion of all inactive outer predecessors; positivity of the Bellman function; and square-summable domain-wall tails whose finite truncations approach q_* . Every computer-assisted item was also reconstructed with a separate `mpmath.iv` implementation and locally written complex intervals, without importing the Arb/FLINT production modules. All gates pass and its enclosure overlaps the production result. The certificate map and technical supplement give every script, margin, failed guard, and the exact independence contract.

7 Scope

Theorem 1 concerns one Bell functional. It does not assert an elementary closed form for q_* , spatiality of arbitrary commuting correlations, or experimental accessibility of the limiting wall. The proof is independent of any broader foundational interpretation. The observed finite-carrier gaps are exponentially consistent with the plateau multiplier, but no quantitative device-independent dimension lower bound is claimed.

Computational disclosure. Frontier language models were used extensively for proof discovery, implementation, adversarial auditing, and editorial assistance. The named author assumes responsibility for the release. Every load-bearing computer-assisted claim is mapped to independently replayable certificates.

Artifacts. The complete sources, frozen receipts, and independent reconstruction are at <https://github.com/Apsiape/i3322-exact-wall>.

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